MCV4U COMPREHENSIVE REVIEW PACKAGE

Vector & Parametric Equations, Planes in $\mathbb{R}^3$, Coplanarity, Distances & Line-Plane Geometry

Course: MCV4U1 (Grade 12 Calculus & Vectors) — Complete worked solutions attached at back.

1. Determine both the vector equation and the set of parametric equations for the plane π that passes through the point $A(2, 3, -1)$ and is parallel to the non-collinear direction vectors $\vec{d_1} = (1, -2, 4)$ and $\vec{d_2} = (3, 0, -1)$.

2. Consider the plane π given by the vector equation $\vec{r} = (4, -1, 2) + s(2, 1, -3) + t(-1, 2, 5)$, $s, t \in \mathbb{R}$.

a) Write the corresponding set of parametric equations for π.

b) Determine algebraically whether the point $Q(7, 3, -7)$ lies on this plane.

3. Find the vector equation of the plane π containing the 3D line $L : \vec{r} = (1, -1, 3) + k(2, 4, -1)$, $k \in \mathbb{R}$, and the external point $P(-2, 1, 5)$.

4. A plane π passes through $P_0(0, 2, -3)$ and is parallel to $\vec{u} = (1, 1, 2)$ and $\vec{v} = (-2, 1, 1)$. Find the parametric equations of the line of intersection between plane π and the xy-coordinate plane ($z = 0$).

5. Determine whether the two vector planes $\pi_1 : \vec{r} = (1, 2, 0) + s(1, -1, 2) + t(2, 1, -1)$ and $\pi_2 : \vec{r} = (3, 0, 1) + p(-1, 2, -3) + q(3, 0, 1)$ are parallel, coincident, or intersecting. Justify your conclusion fully.

6. Express the target vector $\vec{w} = (-5, 17)$ as a linear combination $\vec{w} = a\vec{u} + b\vec{v}$ of the basis vectors $\vec{u} = (3, 1)$ and $\vec{v} = (-2, 4)$ in $\mathbb{R}^2$. Find exact scalar values a and b.

7. Determine whether the vectors $\vec{a} = (2, -1, 3)$, $\vec{b} = (1, 4, -2)$, and $\vec{c} = (5, 7, -1)$ are coplanar in $\mathbb{R}^3$. Show complete vector algebra.

8. Determine if the vector $\vec{x} = (1, 3, 2)$ can be expressed as a linear combination of $\vec{y} = (2, -1, 1)$ and $\vec{z} = (-4, 2, -2)$ in $\mathbb{R}^3$. Explain your result geometrically.

9. Given the vectors $\vec{u} = (1, -2, 2)$, $\vec{v} = (3, 0, -1)$, and $\vec{w} = (k, -4, 5)$, find the value of the scalar parameter $k \in \mathbb{R}$ such that the three vectors are coplanar.

10. Determine the Cartesian (scalar) equation $Ax + By + Cz + D = 0$ of the plane passing through $P(3, -2, 4)$ with normal vector $\vec{n} = (5, 1, -2)$.

11. Find the Cartesian equation of the plane passing through the three non-collinear points $A(2, 1, 0)$, $B(4, -1, 3)$, and $C(1, 3, -2)$. Express your answer with integer coefficients where $A > 0$.

12. Convert the set of parametric equations $x = 1 + 2s - t$, $y = -2 + s + 3t$, $z = 3 - s + 2t$, $s, t \in \mathbb{R}$, into its scalar Cartesian form $Ax + By + Cz + D = 0$.

13. A plane has Cartesian equation $3x - 4y + 2z - 12 = 0$.

a) State the x-, y-, and z-intercepts of the plane.

b) Find two non-collinear direction vectors $\vec{d_1}$ and $\vec{d_2}$ that lie in the plane, and write a vector equation for the plane.

14. Determine the Cartesian equation of the plane that is perpendicular to the line $L : \frac{x-1}{2} = \frac{y+3}{-1} = \frac{z-4}{3}$ and passes through the point $M(-1, 2, 5)$.

15. Calculate the exact distance from the point $P(1, 4)$ to the 2D line $3x - 4y + 7 = 0$.

16. Calculate the exact perpendicular distance from the point $Q(2, -3, 1)$ to the 3D line $L : \vec{r} = (-1, 0, 2) + t(1, 2, -2)$, $t \in \mathbb{R}$. State your answer as an exact simplified radical with a rationalized denominator.

17. Determine the shortest distance from the point $R(-2, 5, 3)$ to the plane $\pi : 2x - y + 2z - 11 = 0$.

18. Calculate the acute angle θ (to the nearest degree) between the two intersecting planes $\pi_1 : 3x - 2y + 6z - 1 = 0$ and $\pi_2 : x + y - 4 = 0$.

19. Determine whether the line $L : \vec{r} = (2, 1, -3) + t(1, -2, 3)$, $t \in \mathbb{R}$, is parallel to, lies completely within, or intersects the plane $\pi : 2x + y - 5 = 0$. If it intersects, find the exact point of intersection.

20. Consider the line $L : \vec{r} = (1, -1, 2) + t(k, 2, -1)$, $t \in \mathbb{R}$, and the plane $\pi : 2x - ky + 4z - 7 = 0$.

a) Solve for all real values of k such that line L is parallel to plane π.

b) For each calculated value of k, determine if line L lies completely within plane π or is strictly parallel to it.

21. Two planes are defined by $\pi_1 : x + ky - z + 4 = 0$ and $\pi_2 : 2x + 3y + mz - 1 = 0$, where $k, m \in \mathbb{R}$.

a) Find the relationship between k and m such that $\pi_1 \perp \pi_2$.

b) Find the exact values of k and m such that $\pi_1 \parallel \pi_2$.

MASTER EXEMPLAR SOLUTIONS

Complete Step-by-Step Model Solutions for Questions 1–21

1. **Given:** Point $A(2,3,-1)$, direction vectors $\vec{d_1} = (1,-2,4)$, $\vec{d_2} = (3,0,-1)$.
Vector Equation: $\vec{r} = (2,3,-1) + s(1,-2,4) + t(3,0,-1),\ s,t \in \mathbb{R}$
Parametric Equations: $\begin{cases} x = 2 + s + 3t \\ y = 3 - 2s \\ z = -1 + 4s - t \end{cases} \quad s,t \in \mathbb{R}$

2. a) Parametric equations for $\vec{r} = (4,-1,2) + s(2,1,-3) + t(-1,2,5)$:
$\begin{cases} x = 4 + 2s - t \\ y = -1 + s + 2t \\ z = 2 - 3s + 5t \end{cases} \quad s,t \in \mathbb{R}$

b) Substitute $Q(7,3,-7)$:
(1) $4 + 2s - t = 7 \implies 2s - t = 3$
(2) $-1 + s + 2t = 3 \implies s + 2t = 4$
From (1), $t = 2s - 3$. Substitute into (2): $s + 2(2s-3) = 4 \implies 5s = 10 \implies s = 2, t = 1$.
Check z: $z = 2 - 3(2) + 5(1) = 2 - 6 + 5 = 1 \neq -7$.
Since $z = 1 \neq -7$, point $Q(7,3,-7)$ does **not** lie on plane π.

3. Point on line $P_0(1,-1,3)$, direction $\vec{d} = (2,4,-1)$, external point $P(-2,1,5)$.
Vector $\vec{P_0P} = P - P_0 = (-2-1, 1-(-1), 5-3) = (-3,2,2)$.
Vector Equation: $\vec{r} = (-2,1,5) + s(2,4,-1) + t(-3,2,2),\ s,t \in \mathbb{R}$.

4. Plane $\pi : \vec{r} = (0,2,-3) + s(1,1,2) + t(-2,1,1)$.
Parametric equations: $x = s - 2t$, $y = 2 + s + t$, $z = -3 + 2s + t$.
Intersection with xy-plane ($z = 0$): $-3 + 2s + t = 0 \implies t = 3 - 2s$.
Substitute t into x and y:
$x = s - 2(3 - 2s) = 5s - 6$
$y = 2 + s + (3 - 2s) = 5 - s$
$z = 0$
Parametric Equations of Intersection Line: $\begin{cases} x = -6 + 5s \\ y = 5 - s \\ z = 0 \end{cases} \quad s \in \mathbb{R}$

5. π_1 normal: $\vec{n}_1 = (1,-1,2) \times (2,1,-1) = (1-2, 4-(-1), 1-(-2)) = (-1,5,3)$.
π_2 normal: $\vec{n}_2 = (-1,2,-3) \times (3,0,1) = (2-0, -3(-3)-(-1)(1), 0-6) = (2,-8,-6) = -2(-1,4,3)$.
Since $\vec{n}_1 = (-1,5,3)$ and $\vec{n}_2 = (2,-8,-6)$ are not parallel (not scalar multiples, as $-1/2 \neq 5/(-8)$), the normals point in different directions.
Conclusion: The two planes are **intersecting** along a line.

6. Set $a(3,1) + b(-2,4) = (-5,17)$:
(1) $3a - 2b = -5$
(2) $a + 4b = 17 \implies a = 17 - 4b$
Substitute a into (1): $3(17-4b) - 2b = -5 \implies 51 - 14b = -5 \implies 14b = 56 \implies b = 4$.
Then $a = 17 - 4(4) = 1$.
Solution: $a = 1, b = 4$ ($\vec{w} = \vec{u} + 4\vec{v}$).

7. Evaluate Triple Scalar Product $\vec{a} \cdot (\vec{b} \times \vec{c})$:

$\vec{b} \times \vec{c} = (1, 4, -2) \times (5, 7, -1) = (-4 - (-14), -10 - (-1), 7 - 20) = (10, -9, -13)$.
$\vec{a} \cdot (\vec{b} \times \vec{c}) = (2)(10) + (-1)(-9) + (3)(-13) = 20 + 9 - 39 = -10 \neq 0$.
Since the triple scalar product is non-zero, the volume spanned is non-zero.
Conclusion: The vectors are **not coplanar**.

8. Notice $\vec{z} = (-4, 2, -2) = -2(2, -1, 1) = -2\vec{y}$. Thus $\vec{y}$ and $\vec{z}$ are collinear and span only a 1D line in $\mathbb{R}^3$.
Any linear combination $a\vec{y} + b\vec{z} = (a - 2b)(2, -1, 1)$ must be a scalar multiple of $(2, -1, 1)$.
For $\vec{x} = (1, 3, 2)$, $(1, 3, 2) \neq c(2, -1, 1)$ since $1/2 \neq 3/(-1)$.
Conclusion: $\vec{x}$ **cannot** be expressed as a linear combination of $\vec{y}$ and $\vec{z}$. Geometrically, $\vec{x}$ lies outside the 1D line spanned by $\vec{y}$ and $\vec{z}$.

9. Set triple scalar product $\vec{w} \cdot (\vec{u} \times \vec{v}) = 0$:
$\vec{u} \times \vec{v} = (1, -2, 2) \times (3, 0, -1) = (2 - 0, 6 - (-1), 0 - (-6)) = (2, 7, 6)$.
$\vec{w} \cdot (\vec{u} \times \vec{v}) = (k)(2) + (-4)(7) + (5)(6) = 2k - 28 + 30 = 2k + 2 = 0 \implies k = -1$.
Solution: $k = -1$.

10. Given $P(3, -2, 4)$, $\vec{n} = (5, 1, -2)$.
Cartesian equation: $5x + y - 2z + D = 0$.
Substitute $P(3, -2, 4)$: $5(3) + (-2) - 2(4) + D = 0 \implies 15 - 2 - 8 + D = 0 \implies D = -5$.
Equation: $5x + y - 2z - 5 = 0$.

11. Points $A(2, 1, 0)$, $B(4, -1, 3)$, $C(1, 3, -2)$.
Vectors: $\vec{AB} = (2, -2, 3)$, $\vec{AC} = (-1, 2, -2)$.
Normal $\vec{n} = \vec{AB} \times \vec{AC} = (4 - 6, -3 - (-4), 4 - 2) = (-2, 1, 2)$.
Use $\vec{n} = (2, -1, -2)$ for $A > 0$: $2x - y - 2z + D = 0$.
Substitute $A(2, 1, 0)$: $2(2) - 1 - 2(0) + D = 0 \implies D = -3$.
Equation: $2x - y - 2z - 3 = 0$.

12. Parametric: $x = 1 + 2s - t$, $y = -2 + s + 3t$, $z = 3 - s + 2t$.
Direction vectors: $\vec{d_1} = (2, 1, -1)$, $\vec{d_2} = (-1, 3, 2)$.
Normal $\vec{n} = \vec{d_1} \times \vec{d_2} = (2 - (-3), 1 - 4, 6 - (-1)) = (5, -3, 7)$.
Plane equation: $5x - 3y + 7z + D = 0$.
Substitute point $(1, -2, 3)$: $5(1) - 3(-2) + 7(3) + D = 0 \implies 5 + 6 + 21 + D = 0 \implies D = -32$.
Equation: $5x - 3y + 7z - 32 = 0$.

13. Plane $3x - 4y + 2z - 12 = 0$.

a) Intercepts:
x-intercept ($y = 0, z = 0$): $3x = 12 \implies x = 4 \implies (4, 0, 0)$
y-intercept ($x = 0, z = 0$): $-4y = 12 \implies y = -3 \implies (0, -3, 0)$
z-intercept ($x = 0, y = 0$): $2z = 12 \implies z = 6 \implies (0, 0, 6)$

b) Direction vectors from intercepts:
$\vec{d_1} = (0, -3, 0) - (4, 0, 0) = (-4, -3, 0)$
$\vec{d_2} = (0, 0, 6) - (4, 0, 0) = (-4, 0, 6)$
Vector Equation: $\vec{r} = (4, 0, 0) + s(-4, -3, 0) + t(-4, 0, 6)$, $s, t \in \mathbb{R}$.

14. Perpendicular to line $L \implies$ line direction $\vec{m} = (2, -1, 3)$ is normal vector $\vec{n} = (2, -1, 3)$.
Plane equation: $2x - y + 3z + D = 0$.
Substitute $M(-1, 2, 5)$: $2(-1) - (2) + 3(5) + D = 0 \implies -2 - 2 + 15 + D = 0 \implies D = -11$.
Equation: $2x - y + 3z - 11 = 0$.

15. Distance from $P(1,4)$ to $3x-4y+7=0$:
$d = \frac{|3(1)-4(4)+7|}{\sqrt{3^2+(-4)^2}} = \frac{|3-16+7|}{\sqrt{25}} = \frac{|-6|}{5} = \frac{6}{5} = 1.2$ units.

16. Distance from $Q(2,-3,1)$ to $L: \vec{r} = (-1,0,2) + t(1,2,-2)$:
Line point $P_0(-1,0,2)$, direction $\vec{m} = (1,2,-2)$, $|\vec{m}| = \sqrt{1+4+4} = 3$.
Vector $\vec{P_0Q} = Q - P_0 = (2-(-1), -3-0, 1-2) = (3,-3,-1)$.
Cross product $\vec{P_0Q} \times \vec{m} = (6-2, -1-(-6), 6-(-3)) = (4,5,9)$.
Magnitude $|\vec{P_0Q} \times \vec{m}| = \sqrt{16+25+81} = \sqrt{122}$.
Distance $d = \frac{|\vec{P_0Q} \times \vec{m}|}{|\vec{m}|} = \frac{\sqrt{122}}{3}$ units.

17. Distance from $R(-2,5,3)$ to plane $2x-y+2z-11=0$:
$d = \frac{|2(-2)-1(5)+2(3)-11|}{\sqrt{2^2+(-1)^2+2^2}} = \frac{|-4-5+6-11|}{\sqrt{9}} = \frac{|-14|}{3} = \frac{14}{3}$ units.

18. Planes $\pi_1: 3x-2y+6z-1=0$ ($\vec{n}_1 = (3,-2,6)$) and $\pi_2: x+y-4=0$ ($\vec{n}_2 = (1,1,0)$).
$|\vec{n}_1| = \sqrt{9+4+36} = 7$, $|\vec{n}_2| = \sqrt{1+1+0} = \sqrt{2}$.
$\vec{n}_1 \cdot \vec{n}_2 = 3(1) + (-2)(1) + 6(0) = 1$.
$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|} = \frac{1}{7\sqrt{2}} = \frac{\sqrt{2}}{14} \approx 0.1010$.
$\theta = \arccos(0.1010) \approx 84.2^\circ \approx 84^\circ$.

19. Line $L: \vec{r} = (2,1,-3) + t(1,-2,3)$, plane $\pi: 2x+y-5=0$ ($\vec{n} = (2,1,0)$).
Test $\vec{m} \cdot \vec{n} = (1)(2) + (-2)(1) + (3)(0) = 2-2+0 = 0$.
Since $\vec{m} \cdot \vec{n} = 0$, line L is parallel to or lies within plane π.
Test point $P_0(2,1,-3)$ in plane equation: $2(2) + (1) - 5 = 4+1-5 = 0$.
Since P_0 satisfies the plane equation, line L **lies completely within** plane π.

20. Line direction $\vec{m} = (k,2,-1)$, plane normal $\vec{n} = (2,-k,4)$.

a) $L \parallel \pi \iff \vec{m} \cdot \vec{n} = 0$:
$(k)(2) + (2)(-k) + (-1)(4) = 0 \implies 2k - 2k - 4 = 0 \implies -4 = 0$, which is impossible.
Solution: There are **no real values of** k such that line L is parallel to plane π.

b) Since L is never parallel to π, for all $k \in \mathbb{R}$, line L intersects plane π at a single point (it is never strictly parallel nor contained within).

21. Planes $\pi_1: x+ky-z+4=0$ ($\vec{n}_1 = (1,k,-1)$), $\pi_2: 2x+3y+mz-1=0$ ($\vec{n}_2 = (2,3,m)$).

a) $\pi_1 \perp \pi_2 \iff \vec{n}_1 \cdot \vec{n}_2 = 0$:
$1(2) + k(3) + (-1)(m) = 0 \implies 2 + 3k - m = 0 \implies m = 3k+2$.

b) $\pi_1 \parallel \pi_2 \iff \vec{n}_1 = c\vec{n}_2$:
$\frac{1}{2} = \frac{k}{3} = \frac{-1}{m} \implies k = \frac{3}{2}, m = -2$.